MAgent Theory 2.0

A Physicist's Companion

Robert C. Dennis

Independent Researcher

October 2026

Contents

1 What an Agent Carries **1**
1.1 Beliefs, models, and predictions . . . 1
1.2 A record has an address inside the agent . . . 2
1.3 Which probability space is being used? . . . 2

2 Frames, Messages, and Interactions **3**
2.1 Reading a link from source to receiver . . . 3
2.2 Changing coordinates . . . 4
2.3 Selective interfaces and an explicit pair . . . 4

3 From Statistical Change to Base Geometry **5**
3.1 Relating neighboring contexts . . . 5
3.2 Curvature and paths . . . 6
3.3 Local descriptions and global compatibility . . . 6
3.4 Distinguishability supplies a fiber metric . . . 7
3.5 Pulling distinguishability onto the context base . . . 7
3.6 Choosing a collective connection . . . 8
3.7 What compression does to this geometry . . . 9
3.8 Geometric structure and physical interpretation . . . 10

4 Coupled Relaxation and Source Selection **11**
4.1 The potential and its source rows . . . 11
4.2 Every interaction role contributes to motion . . . 12
4.3 A record likelihood defines a different objective . . . 13

5 What a Message Lets the Receiver Infer **14**
5.1 Selective sending and conditional reconstruction . . . 14
5.2 Agreement needs a common prediction rule . . . 15
5.3 Compression should name what matters . . . 16

6 When a Collective Can Act as an Agent **17**
6.1 Two routes to the same parent . . . 17
6.2 When a gradient flow descends . . . 17
6.3 A missing role can prevent closure . . . 18
6.4 An exact supplied hierarchy . . . 19
6.5 Memory and the remaining formation problem . . . 19

7 A Nonsmooth Route to Recurrent Formation **21**
7.1 Laws, a model, and inherited mass . . . 21
7.2 The interaction that permits contact . . . 22
7.3 Two successive formation events . . . 23
7.4 What the parent retains . . . 23

8 Keeping Distinct Children Under Smooth Dynamics **25**
8.1 A hidden conditional law can move a visible marginal . . . 25
8.2 Retain the distribution of density values . . . 25
8.3 What a future record is allowed to inspect . . . 26
8.4 A smooth channel can erase a dependence in finite time . . . 27

9 A collective trajectory is not a complete interface **28**

10 A Parent That Keeps Its Responses **30**
10.1 Four children and three public circles . . . 30
10.2 Why the original dynamics closes . . . 31
10.3 What the inherited interface permits . . . 32

A Notation in Physical Terms **33**

Chapter 1

What an Agent Carries

A classical statistical description distinguishes a phase-space point from a distribution over phase space. The same distinction begins this theory: $k \in \mathsf{K}$ is a represented possibility, while q is a probability law over such possibilities. The agent carries the law and mechanisms for changing it. Its contents need not be positions and momenta, and the statistical law need not describe an ensemble of physically separate agents.

1.1 Beliefs, models, and predictions

There are two sectors. The belief law q_i describes state contents k. The model law s_i describes model presentations $m \in \mathsf{M}$. Their most general joint description is

$$\zeta_i(dk, dm) = q_i(dk \mid m) s_i(dm). \tag{1.1}$$

The notation $q_i(dk \mid m)$ means a conditional law, and $q_i(dk)$ without the condition denotes its marginal. Choosing $\zeta_i = q_i \otimes s_i$ removes dependence between the sectors in that description. It still permits coupled equations for q_i and s_i, just as coupled equations for two coordinates do not require a statistical ensemble over those coordinates.

A model presentation becomes predictive through a normalized evaluator $G_i(m, dk)$. For each m, this assigns a probability law to the state contents. Averaging the evaluator over the model law gives

$$(s_i G_i)(A) = \int G_i(m, A) s_i(dm). \tag{1.2}$$

For example, let k, m be binary, with $s_i(m = 1) = 0.6$, $G_i(1, \{1\}) = 0.9$, and $G_i(0, \{1\}) = 0.2$. The model mixture assigns probability 0.62 to $k = 1$. A present belief $q_i(k = 1) = 0.8$ is a different object. An interaction rule can bring these descriptions into agreement, revise the model law, or change both. Those dynamics must be specified.

An evaluator is a Markov kernel: a function whose output is a probability measure. Ordinary functions are included as the case in which the output measure is concentrated at one value. This language accommodates discrete, continuous, singular, and multimodal laws without selecting a Gaussian family. When differentiating laws later, we will restrict to a regular statistical family or positive finite tables.

1.2 A record has an address inside the agent

Let $x_i \in \mathsf{S}_i$ denote the complete internal state, including its laws, interfaces, and any memory. A record readout is an ordinary measurable function

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i. \tag{1.3}$$

The output alphabet O_i specifies the distinctions the agent can read. A binary register gives $\mathsf{O}_i = \{0, 1\}$; a real-valued comparison can use $\mathbb{R}^r$. The detector analogy is precise at this level: an internal state contains a readable distinction. It does not require an outside observer to inspect it.

Write $x \in \mathsf{S}$ for the complete population configuration. A deterministic interaction $F_{ij} : \mathsf{S} \to \mathsf{S}$ produces the record

$$o_{ij} = \rho_i\big(\mathrm{pr}_i\, F_{ij}(x)\big), \tag{1.4}$$

where pr_i selects the receiver's state. For a stochastic transition $\Gamma_{ij}(x, dx')$, the probability of a record in A is

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\big(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\big). \tag{1.5}$$

This is a pushforward: collect the probability of every next state that displays an outcome in A. The transition specifies what happens; the readout specifies what is accessible. A persistent record requires that some later readout still recover the distinction. A transient comparison and a memory trace therefore have different dynamical requirements.

A likelihood on represented contents has another role. For instance, the expression $o = A_i k_i - B_{ij} k_j + \epsilon$ gives a comparison value once the contents and noise are given. Their joint law predicts possible comparisons. A registration rule must still place the outcome in an internal state and specify the response to it.

1.3 Which probability space is being used?

The configuration x contains the agents' laws. A tuple of represented contents $(k_i, m_i)_i$ belongs to a different space. A probability law over possible configurations is a third construction, useful when a preparation or stochastic dynamics has been declared. Deterministic motion of x requires neither sampling the contents nor drawing an ensemble of configurations at each step.

This distinction also clarifies correlations. Two independent fair bits have probability $1/4$ for each pair. Condition on a shared record that the bits agree. Their marginals remain fair, but the conditional joint law has probability $1/2$ on each of $(0, 0)$ and $(1, 1)$. A calculation of the next joint comparison needs this dependence; the marginals do not contain it. A coupled product-family evolution remains a possible model, but it will not represent that exact conditional joint law. Whether a correlation is needed is decided by the response being predicted.

The laws can be attached to local domains of a context base $\mathcal{C}$. This is the section picture: a context selects the law applicable there, and a frame specifies its coordinates. Most interactions below occur at a common context. Moving between contexts introduces the connection problem in chapter 3.

Chapter 2

Frames, Messages, and Interactions

An agent's probability law and the coordinates used to describe that law are different objects. Communication introduces a third object: a rule for translating, filtering, or modifying what another agent supplies. Keeping these distinctions explicit prevents a coordinate change from being mistaken for an interaction, or a selective message from being mistaken for an invertible transport.

2.1 Reading a link from source to receiver

There are two sectors. Write $\nu_i^q = q_i$ for the belief marginal and $\nu_i^s = s_i$ for the model marginal. In a finite-dimensional realization, a sector $a \in \{q, s\}$ has a reference vector space V_a, an agent space E_i^a, and an invertible presentation

$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{\nu}_i^a = (U_i^a)_{\#}^{-1} \nu_i^a. \tag{2.1}$$

The pushforward notation expresses the represented possibilities in reference coordinates. The two sectors may use representations of the same frame group while having different dimensions, laws, and interaction rules.

A law-dependent reference generator and its displayed link are

$$L_{ij}^a = \mathcal{L}_{ij}^a(\bar{\nu}_i^a, \bar{\nu}_j^a), \qquad \Theta_{ij}^a = U_i^a e^{L_{ij}^a} (U_j^a)^{-1}. \tag{2.2}$$

The index order is receiver first, source second. Reading from right to left, the source is expressed in reference coordinates, the generator acts there, and the result is expressed at the receiver. The function $\mathcal{L}_{ij}^a$ is a constitutive choice. The exponential parameterization does not derive it or equate the belief and model generators.

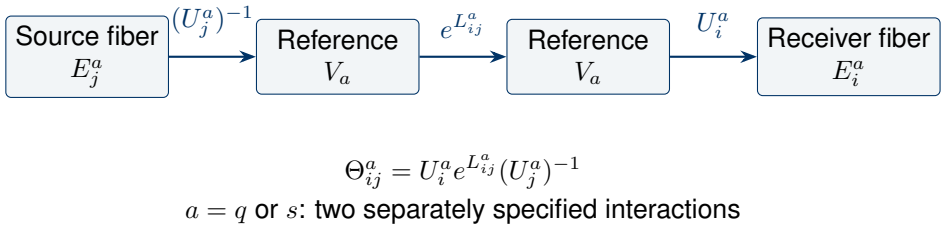


Figure 2.1: The source presentation is undone by $(U_j^a)^{-1}$, the interaction $e^{L_{ij}^a}$ acts within the reference space, and U_i^a presents the result at the receiver. Endpoint frames and the intervening interaction have separate roles.

2.2 Changing coordinates

For a passive change $y_i' = g_i^a y_i$, both the law and its presentation change: $\nu_i^{a\prime} = (g_i^a)_{\#}\nu_i^a$ and $U_i^{a\prime} = g_i^a U_i^a$. The reference law stays fixed, giving

$$\Theta_{ij}^{a\prime} = g_i^a \Theta_{ij}^a (g_j^a)^{-1}. \tag{2.3}$$

This endpoint covariance is familiar from gauge link variables [14]. Metrics and additional model data must also be re-expressed when forming scalar predictions. Covariance alone supplies no physical symmetry or evolution law.

Changing the common reference presentation is a separate operation. Under $U_i^a \mapsto U_i^a h_a$, the reference laws become $(h_a^{-1})_{\#}\bar{\nu}_i^a$. Reference consistency requires the constitutive rule, including any interface data, to give $L_{ij}^{a\prime} = h_a^{-1} L_{ij}^a h_a$. The identity $e^{h_a^{-1} L h_a} = h_a^{-1} e^L h_a$ then leaves the displayed link unchanged.

2.3 Selective interfaces and an explicit pair

Sending and receiving can be represented by

$$T_j^a : V_a \longrightarrow H_a, \qquad O_i^a : H_a \longrightarrow V_a, \qquad L_{ij}^a = O_i^a T_j^a. \tag{2.4}$$

The sender T_j^a extracts a message; the receiver O_i^a inserts its effect. A direct content map $O_i^a T_j^a$ can lose dimensions. Its exponential is always invertible and acts as identity on $\ker T_j^a$. Thus exponentiating a selective generator retains unread directions. Sending only the extracted message is a different operation. General normalized Markov kernels allow communication without any linear realization.

One explicit law-dependent choice uses finite second moments and positive-definite covariance. Let μ_i^a, M_i^a be the reference mean and covariance. On nonself edges choose a scalar λ_{ij}^a and set

$$L_{ij}^a = \lambda_{ij}^a \mu_i^a (\mu_j^a)^T (M_j^a)^{-1}. \tag{2.5}$$

The row $(\mu_j^a)^T (M_j^a)^{-1}$ sends a scalar message; the column $\lambda_{ij}^a \mu_i^a$ receives it. Under a reference change, the mean transforms by h_a^{-1} and covariance by congruence. Their product consequently transforms by conjugation. These assumptions concern moments, not Gaussian shape. Primitive self-comparisons are set to identity separately.

For a concrete non-Gaussian example, let Z have independent centered Laplace coordinates of variance one in two dimensions. Give two agents the laws of $e_1 + Z$ and $e_2 + Z$, with identity frames and unit coefficients. Then $M_1 = M_2 = I$, $L_{12} = E_{12}$, and $L_{21} = E_{21}$. Since each generator squares to zero,

$$\Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.6}$$

Here E_{ab} has its only nonzero entry, one, in row a, column b. The directed communication loop is nontrivial. It establishes neither a connection over contexts nor a physical curvature.

Chapter 3

From Statistical Change to Base Geometry

The agent graph and the context base describe different directions of comparison. An edge $j \to i$ compares agents at one context. A path in the base $\mathcal{C}$ compares states at successive contexts. A product of communication links can be nontrivial without defining curvature on $\mathcal{C}$. Conversely, one agent can carry a curved connection over contexts; collective averaging is not required.

3.1 Relating neighboring contexts

To construct that connection, suppress the sector label and fix one agent. On a coordinate neighborhood, take a twice continuously differentiable presentation $U(c)$ and a continuously differentiable matrix-valued one-form $a = a_\mu(c)dc^\mu$. Declare directional links whose first-order behavior is

$$\Theta(c + \delta c, c) = U(c + \delta c)e^{-a_\mu(c)\delta c^\mu}U(c)^{-1} + O(|\delta c|^2). \tag{3.1}$$

The displacement enters linearly at first order, and zero displacement gives identity. These are additional properties beyond having a finite interaction matrix at each context. For a principal G-connection, the generators must belong to the represented Lie algebra; unrestricted matrices instead define general linear bundle connections.

Using $D = d + B$, expansion identifies

$$B = UaU^{-1} - dUU^{-1}. \tag{3.2}$$

The first term carries the declared directional interaction into displayed coordinates. The second compensates for the change of presentation along the path. Their relationship is exactly the operator identity

$$d + B = U(d + a)U^{-1}. \tag{3.3}$$

Indeed, acting on a vector field f gives $Ud(U^{-1}f) = df - dUU^{-1}f$; the remaining term is $UaU^{-1}f$. Squaring the operator therefore gives

$$F = dB + B \wedge B = U(da + a \wedge a)U^{-1}. \tag{3.4}$$

This is the mathematical connection to gauge theory, with the sign convention fixed by the displayed covariant derivative [13].

3.2 Curvature and paths

Curvature need not come from spatially varying coefficients. On a two-dimensional context patch choose $U = I$, $a_x = E_{12}$, and $a_y = E_{21}$. Both directional generators have rank one, but

$$F_{xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.5}$$

The matrix E_{ab} has one unit entry in position (a, b). Applying the two infinitesimal operations in opposite orders gives different results. This example supplies directional data; it does not derive them from the agents' laws. A presentation change alone, $B = -dUU^{-1}$, instead has zero curvature.

Finite path transport solves

$$\frac{dT}{dt} = -B(\dot{\gamma}(t))T(t), \qquad T(0) = I. \tag{3.6}$$

Its solution is a path-ordered exponential. Concatenating paths composes their transports; reversing the same path gives the inverse. Two independently specified attending edges can have unrelated generators, so their opposite labels do not impose this inverse relation. The distinction is necessary when borrowing Wilson-loop language from gauge theory [14].

3.3 Local descriptions and global compatibility

Endpoint covariance requires

$$\Theta'(c_2, c_1) = g(c_2)\Theta(c_2, c_1)g(c_1)^{-1}. \tag{3.7}$$

Differentiating the endpoints produces the affine law

$$B' = gBg^{-1} - dgg^{-1}. \tag{3.8}$$

Likewise, keeping the intrinsic transport fixed under $U' = Uh$ requires $a' = h^{-1}ah + h^{-1}dh$. A directional receiving–sending product that transforms only by conjugation is a tensorial one-form. It can supply the difference between two connections, but does not by itself supply the full affine transformation law.

Global compatibility adds another requirement. If patch coordinates obey $v_\alpha = g_{\alpha\beta}v_\beta$, then $g_{\alpha\beta}g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps, and local connection forms must satisfy the affine law there. A nontrivial bundle may have no single global presentation. Even a trivial bundle can carry flat transport with global memory: on a circle, the scalar connection $B = bd\theta$ has $F = 0$ but holonomy $e^{-2\pi b}$ around one turn. For real $b \neq 0$, this differs from identity.

3.4 Distinguishability supplies a fiber metric

A connection tells us how to compare laws at neighboring contexts. To measure the change remaining after that comparison, use the Fisher metric on a declared regular statistical family. Let ζ_θ be a family of probability laws, with a centered square-integrable score ℓ_v in parameter direction v. In a dominated differentiable realization,

$$\ell_v = \left.\frac{d}{d\epsilon}\right|_{\epsilon=0} \log p_{\theta+\epsilon v}, \qquad g^F_\zeta(v,w) = \mathbb{E}_\zeta[\ell_v \ell_w]. \tag{3.9}$$

Quadratic-mean differentiability supplies a more general score definition. Positive definiteness requires that the admitted tangent directions be statistically distinguishable. These hypotheses apply to the chosen family, not to every element of the measurable space of laws [2].

The relevant law is the full $\zeta(dk, dm) = q(dk \mid m)s(dm)$. Its log density has a marginal part and a conditional part. The conditional score averages to zero for each m, so the cross term disappears:

$$g^F_\zeta = g^F_s + \int g^F_{q(\cdot|m)} s(dm). \tag{3.10}$$

For the product family this is $g^F_q + g^F_s$. For a general joint family the second term is conditional geometry. It cannot be replaced by the geometry of the state marginal.

For example, take $k, m \in \{-1, 1\}$ and

$$\zeta_\rho(k,m) = \frac{1+\rho km}{4}, \qquad |\rho| < 1. \tag{3.11}$$

Both marginals are uniform for every ρ. The score is $km/(1+\rho km)$, giving

$$g^F_\zeta(\partial_\rho, \partial_\rho) = \frac{1}{2}\left(\frac{1}{1+\rho} + \frac{1}{1-\rho}\right) = \frac{1}{1-\rho^2}. \tag{3.12}$$

The law changes through dependence even though neither marginal changes. This is why the statistical metric must follow the actual probabilistic description.

3.5 Pulling distinguishability onto the context base

Let $\zeta(c)$ be an admitted smooth section and let ω be its connection. Subtract the horizontal motion from the derivative of the section:

$$D^\omega\zeta = \mathrm{ver}^\omega\, T\zeta, \qquad h^\omega_\zeta(X,Y) = g^F_\zeta(D^\omega\zeta X, D^\omega\zeta Y). \tag{3.13}$$

The tangent vectors X, Y now lie in the context base. The tensor h^ω_ζ measures how much the represented law changes when the context changes, after the declared transport accounts for the change of frame. In a local chart it uses $D^A\zeta = d\zeta + J_\zeta A$, where $J_\zeta\xi$ is the infinitesimal action of the Lie algebra element ξ on the law.

Under second-order relative-entropy regularity, move a small distance ϵX, transport the neighboring law back, and compare it with the original:

$$D_{\mathrm{KL}}\left(\zeta(c)\middle\|T^{\omega}_{c+\epsilon X\to c\#}\zeta(c+\epsilon X)\right)=\frac{\epsilon^2}{2}h^{\omega}_{\zeta}(X,X)+o(\epsilon^2). \tag{3.14}$$

The familiar local quadratic behavior of KL therefore measures contextual distinguishability. The transport is part of the comparison.

For a simple case with identity transport, let the law at real context c be $\mathrm{Ber}(p(c))$, with $0<p(c)<1$. Direct differentiation of its two log probabilities gives

$$h=\frac{p'(c)^2}{p(c)(1-p(c))}dc^2. \tag{3.15}$$

The same change of success probability is more distinguishable near a deterministic outcome. If $p'(c)=0$, this family records no first-order distinction in that context direction. The example uses no Gaussian assumption.

A passive gauge change transforms both the section and connection, leaving the tensor unchanged. Choosing a different connection changes what is subtracted. If A changes by a Lie-algebra-valued one-form α, the statistical derivative changes by $J_{\zeta}\alpha$; its squared Fisher norm acquires both cross terms and the squared norm of that added vector. Gauge invariance is thus a statement about presentation, not a claim that all transport rules give the same geometry.

Positive-definite fiber geometry makes $\ker h^{\omega}_{\zeta}=\ker D^{\omega}\zeta$. Constant rank gives a metric on the vector spaces obtained by quotienting these null directions. To obtain a lower-dimensional context manifold, the null directions must also fit together into leaves and the metric must descend along them. The one-form $dz-xdy$ illustrates the difference: its square has rank one everywhere, but its kernel is not integrable because $(dz-xdy)\wedge d(dz-xdy)$ is nonzero. Counting statistical directions does not by itself construct a quotient space of contexts.

3.6 Choosing a collective connection

Suppose several laws have been placed in compatible fibers by declared comparison maps. Give their product tangent space the positive metric

$$\mathsf{G}=\sum_i w_i g^{F}_{p_i},\qquad w_i>0. \tag{3.16}$$

The weights specify how the individual statistical changes are compared. A common infinitesimal frame motion ξ changes the tuple by $J_{\boldsymbol{p}}\xi$. For each contextual direction X, select the correction a that minimizes

$$\|d\boldsymbol{p}(X)+J_{\boldsymbol{p}}a\|^2_{\mathsf{G}}. \tag{3.17}$$

Differentiation with respect to a gives the normal equation

$$\mathsf{G}(JA^{F}(X),J\xi)=-\mathsf{G}(d\boldsymbol{p}(X),J\xi)\quad\text{for every }\xi. \tag{3.18}$$

The residual is orthogonal to all common frame motions. This is the statistical version of the orthogonal connection used to separate rotation from shape change [5].

For an explicit calculation, take two unit-variance normal location laws with means $\mu_1 = m + d$ and $\mu_2 = m - d$, and give them equal weight. Common translation acts as $Ja = (a, a)$. The cost is

$$(dm + dd + a)^2 + (dm - dd + a)^2 = 2(dm + a)^2 + 2(dd)^2. \tag{3.19}$$

It is minimized at $A^F = -dm$, leaving

$$h^{\min} = 2(dd)^2. \tag{3.20}$$

Common location change is removed while change in their relative separation remains. This calculation is a realization of the general construction; the construction itself applies to any admitted smooth isometric group action.

If some nonzero ξ moves none of the laws, then $J\xi = 0$ and the normal equation cannot determine that connection component. At constant orbit rank it determines a smooth affine class modulo this infinitesimal stabilizer. When the infinitesimal stabilizer is zero the solution is unique and obeys the affine patch law. A free proper action additionally gives the usual smooth statistical orbit space. These conclusions concern different hypotheses [9].

Orthogonality also measures the cost of choosing another connection:

$$h^A = h^{\min} + \mathfrak{k}(A - A^F, A - A^F), \qquad \mathfrak{k}(\xi, \eta) = \mathsf{G}(J\xi, J\eta). \tag{3.21}$$

The minimized tensor is positive semidefinite. Whether it has enough independent directions, and whether inference preserves those directions, depends on the actual section and its dynamics. The least-squares rule alone supplies neither persistence nor a physical interpretation of the contextual base.

3.7 What compression does to this geometry

A fixed normalized channel K transforms a family P_θ into $P_\theta K$. The output score is the input score averaged over possibilities compatible with the output. The law of total covariance gives

$$I(P_\theta) - I(P_\theta K) = \mathbb{E}_\theta \operatorname{Cov}_\theta(\ell_\theta(X) \mid Y). \tag{3.22}$$

The missing Fisher information is exactly the uncertainty remaining about the score. A zero value recovers that local score, not necessarily the full family of laws [10]. If the channel itself varies with θ, its own score must be added before conditioning. The output can then acquire information through the channel parameter, so contraction relative to the input alone need not hold.

Even a fixed equivariant channel need not preserve the collectively selected connection. In the two-location example, take $(\mu_1, \mu_2) = (x, -x)$. The fine metric is $\operatorname{diag}(1, 1)$ and its collective connection is zero. Add independent unit-variance normal noise to the first observation. The first variance

becomes two, so the output Fisher metric is $\mathrm{diag}(1/2, 1)$. Its connection minimizes

$$\frac{1}{2}(1+a)^2 + (-1+a)^2, \tag{3.23}$$

whose derivative is $3a-1$. The new minimum is $a = 1/3$. The channel has reduced distinguishability and commutes with translation, yet the weighting of shape change against translation has changed.

In general, let a smooth equivariant retention R map the fine family to the coarse family, with metrics g, g_c. Set $\Delta = g - R^* g_c$, and let r be the fine horizontal residual. Because $dRJ = J^c$, the fine connection also minimizes the coarse cost exactly when

$$\Delta(r(X), J\xi) = 0 \quad \text{for every } X, \xi. \tag{3.24}$$

This follows by inserting the fine connection in the coarse normal equation and using fine orthogonality. It is a condition on the mixed shape–orbit part of information loss. Preserving all orbit scores is sufficient for a fixed channel, but contraction alone is not.

For independently chosen fine and coarse connections there is another contribution. A bundle map may send a fine horizontal vector to a vector that is not coarse-horizontal. If a denotes that vertical difference and $u = D^\omega \zeta$, the coarse derivative is $T^V \Psi u + a$. Expanding its squared norm gives the fine information loss minus two cross terms and minus $\|a\|^2$. Zero horizontal mismatch is sufficient for contraction of the base tensor; a nonzero mismatch can still be dominated by information loss. The actual derivative and both connections must therefore be specified.

3.8 Geometric structure and physical interpretation

The pullback construction is already a mathematical geometry of contextual statistical change. Its physical interpretation requires a model of the contexts, a mechanism selecting the comparison data, and observable responses to transport. Inherited evolution and inherited responses impose further conditions on a parent; connection preservation is an additional requirement.

Fisher geometry is positive semidefinite. An indefinite metric can be introduced through a separate construction. For example, a Hermitian 2×2 matrix $H = tI + \boldsymbol{x} \cdot \boldsymbol{\sigma}$ has determinant $t^2 - |\boldsymbol{x}|^2$, where $\boldsymbol{\sigma}$ are the Pauli matrices. Pulling its determinant bilinear form back by an invertible soldering field gives a Lorentzian metric. A circle extension then requires a circle connection and a positive radius. Those choices do not follow from Fisher positivity, and their existence does not establish spacetime dynamics.

Chapter 4

Coupled Relaxation and Source Selection

The dynamical variables are probability laws. Moving an agent therefore means changing its expectations, uncertainties, and dependencies, rather than moving a point through its sample space. The physical analogue developed here is overdamped motion on a probability manifold: an interaction potential supplies the driving covector, and the Fisher metric converts that covector into a velocity. Neither ingredient determines the other.

4.1 The potential and its source rows

At a fixed context, let q_i describe agent i's represented state k, and let s_i describe its model presentation m. For this construction their joint recognition law is the explicit product $q_i \otimes s_i$. Choose a model reference $r_i(dm)$ and a normalized evaluator $G_i(m, dk)$. Their local contribution is

$$\mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| r_i G_i) = \sum_i \left[D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot)) s_i(dm) \right]. \tag{4.1}$$

The equality separates the model marginal from the conditional state law. Thus a product recognition family still has coupled dynamics: changing s_i changes the evaluator against which q_i relaxes, and changing q_i changes the model costs.

The channels of chapter 2 send peer laws into comparable spaces. Write $\nu^q_{ij} = q_j \mathcal{K}^q_{ij}[\mathbf{q}]$ and $D^q_{ij} = D_{\mathrm{KL}}(q_i \| \nu^q_{ij})$, where $\mathbf{q}$ collects the belief laws. Define ν^s_{ij} and D^s_{ij} similarly for model laws. With normalized source rows β_i, γ_i, positive reference rows π^q_i, π^s_i, and $T_q, T_s > 0$, the potential is

$$\begin{aligned} \mathcal{E} = \mathcal{E}_0 &+ \sum_i \left[\sum_j \beta_{ij} D^q_{ij} + T_q D_{\mathrm{KL}}(\beta_i \| \pi^q_i) \right] \\ &+ \sum_i \left[\sum_j \gamma_{ij} D^s_{ij} + T_s D_{\mathrm{KL}}(\gamma_i \| \pi^s_i) \right]. \end{aligned} \tag{4.2}$$

Each sum uses its declared source set. The row penalty balances preference for compatible peers against departure from the reference routing.

Hold the laws fixed. For either sector, minimize $f(\eta) = \sum_j \eta_j D_j + T D_{\mathrm{KL}}(\eta\|\pi)$ subject to $\sum_j \eta_j = 1$. Stationarity gives $D_j + T[\log(\eta_j/\pi_j) + 1] = \lambda$, hence

$$\eta_j^* = \frac{\pi_j e^{-D_j/T}}{Z}, \qquad Z = \sum_j \pi_j e^{-D_j/T}, \qquad f(\eta) = -T \log Z + T D_{\mathrm{KL}}(\eta\|\eta^*). \tag{4.3}$$

The last identity proves that this stationary row is the unique minimum. Eliminating both sets of rows gives $\mathcal{E}_{\mathrm{red}} = \mathcal{E}_0 - T_q \sum_i \log Z_i^q - T_s \sum_i \log Z_i^s$. At fixed temperatures and reference rows,

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta_{ij}^* dD_{ij}^q + \sum_{ij} \gamma_{ij}^* dD_{ij}^s. \tag{4.4}$$

This follows by differentiating each log partition sum. Optimized row derivatives disappear; derivatives inside their costs remain.

4.2 Every interaction role contributes to motion

For normalized densities a, b, variation of $\int a \log(a/b)$ gives

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b) da - \int (a/b) db, \qquad d\nu_{ij}^q = (dq_j)\mathcal{K}_{ij}^q + q_j D\mathcal{K}_{ij}^q(d\mathbf{q}). \tag{4.5}$$

Here da, db denote mass-preserving variations, and differentiation assumes a regular dominated family. An agent moves through its receiver role, its source role, and any dependence of the channels on its law. Directed routing does not remove incoming contributions to source motion. For an identity self-channel, the two laws remain identical and their divergence stays zero. Holding an old self snapshot fixed instead creates a different potential after the current law changes.

For positive finite probability tables, collect all dynamical tables as u^a, including the source rows when they are not eliminated. The Fisher metric and its unit-mobility descent are

$$\begin{aligned} g_u(v,v) &= \sum_{a,z} \frac{(v_z^a)^2}{u_z^a}, \\ \dot{u}_z^a &= -u_z^a(h_z^a - \mathbb{E}_{u^a} h^a), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}, \\ \dot{\mathcal{E}} &= -\sum_a \mathrm{Var}_{u^a}(h^a). \end{aligned} \tag{4.6}$$

Centering preserves each normalization. Substituting the velocity into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ proves the dissipation identity. The same construction uses the Fisher tensor on a regular smooth family [1]. Both sectors have equal mobility; a slower model response must arise from the resulting dynamics.

A Bernoulli law makes the metric concrete. Its success probability p has metric $dp^2/[p(1-p)]$, so the mobility is $p(1-p)$. For $\mathcal{E}(p) = D_{\mathrm{KL}}(\mathrm{Ber}(p)\|\,\mathrm{Ber}(p_0))$, with $p, p_0 \in (0,1)$, write $l = \log[p/(1-p)]$ and $l_0 = \log[p_0/(1-p_0)]$. Then

$$\dot{p} = -p(1-p)(l - l_0), \qquad \dot{l} = -(l - l_0). \tag{4.7}$$

Log odds relax exponentially, although probability does not follow a linear restoring equation. Here t parametrizes the chosen relaxation, and the dimensionless T_q, T_s weight routing penalties. They have not been identified with physical time or bath temperature.

4.3 A record likelihood defines a different objective

A fixed normalized prior P_0 and record likelihood ℓ_o give posterior $\Pi_o = \ell_o P_0 / Z_o$ and the exact identity

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o. \tag{4.8}$$

Source inference in a product recognition family then uses expected negative log likelihood as its cost. Peer coupling instead uses divergence between laws, whose source template can move during relaxation. Both yield Gibbs rows because both contain the same categorical relative-entropy penalty; their costs and state derivatives differ. Identical independent Gaussian laws have zero peer divergence but nonzero expected squared sample difference. A normalized likelihood for that difference therefore describes an interaction distinct from direct agreement of the laws.

Chapter 5

What a Message Lets the Receiver Infer

Sending a few observables can change an agent's description of many variables. The extra structure comes from the receiver's conditional model. This is familiar in physics: measuring one coordinate of a correlated system can change predictions for another coordinate without measuring it directly. The distinction between transmitted information and inferred structure is explicit here.

5.1 Selective sending and conditional reconstruction

A sender uses a measurable map $T_j : E_j \to \mathsf{H}$ into a message space. If its content has law q_j, possible messages have law

$$\nu_j = (T_j)_{\#} q_j. \tag{5.1}$$

The pushforward symbol means $\nu_j(A) = q_j(T_j^{-1}A)$. Sampling a message and transmitting the whole law ν_j are different communication protocols. A linear receiving map $O_i : \mathsf{H} \to E_i$ embeds a message, but its composition with T_j cannot have higher rank than T_j.

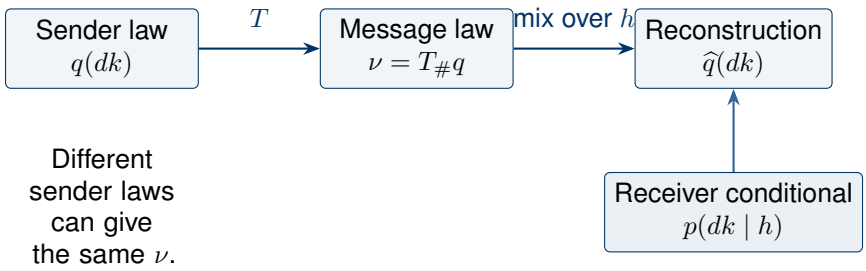


Figure 5.1: Selective communication and receiver structure. The sender fixes the transmitted marginal. The receiver supplies a conditional law for distinctions that the message does not specify. This distributional reconstruction presupposes that the message law is available; one received sample supplies a value instead.

To see exactly what is lost, work in one common presentation and write $T : E \to \mathsf{H}$. Let q be the sender's law and p the receiver's reference law, with finite $D_{\mathrm{KL}}(q\|p)$. The receiver can combine the transmitted marginal $\nu = T_{\#}q$ with its own conditional law:

$$\widehat{q}(dk) = \int p(dk \mid h)\nu(dh). \tag{5.2}$$

Disintegration on standard Borel spaces provides these conditional laws [7]. The reconstruction has exactly the right message marginal. Its remaining discrepancy is

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\nu(dh). \tag{5.3}$$

Both laws assign the same probability to h, so that factor cancels in their density ratio. The remaining logarithm compares the two conditional laws, which gives the equality by integration.

For a calculation on four points, let $k = (H, V)$ be two bits. The sender gives probability $1/2$ to each of 00 and 11, and transmits only H. A receiver whose reference makes both bits independent and fair reconstructs the uniform law on all four points. The message marginal is correct, while the relative entropy is $\log 2$. A receiver whose conditional model has $V = H$ reconstructs the sender's full law. The missing correlation was supplied by the receiver.

One reference conditional can recover a whole family only under a stronger condition. Suppose $q_\alpha = L_\alpha p$ and use the same $p(dk \mid h)$ for every α. Its reconstruction has density

$$\frac{d\widehat{q}_\alpha}{dp}(k) = \mathbb{E}_p[L_\alpha \mid T](k). \tag{5.4}$$

To obtain this identity, weight the reference message marginal by its conditional mean likelihood and then apply the reference conditional. The result equals q_α precisely when L_α is already measurable from the message. Reconstructing one posterior with its own conditional does not establish this family-wide property. If a local reconstruction must also ignore retained exterior information, the reference needs conditional independence from that exterior given the message.

For a single received value h, Bayesian assimilation instead takes the form

$$q_i^+(dk) = \frac{\ell_i(h \mid k) q_i^-(dk)}{\int \ell_i(h \mid k') q_i^-(dk')}, \tag{5.5}$$

with a positive finite denominator. In the perfectly correlated two-bit example, a noiseless observation $H = 1$ also fixes $V = 1$. This is inference through an existing relation, not an inversion of an arbitrary lossy map.

5.2 Agreement needs a common prediction rule

Suppose a model law s and a normalized record evaluator $K(m, do)$ predict the record law sK. If two agents compare model laws in one frame, they must also compare their evaluators in that frame. Equal model laws can predict opposite bits when their evaluators differ.

For one common evaluator and finite $D_{\mathrm{KL}}(s\|t)$, an exact information balance is

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do), \tag{5.6}$$

where s^o, t^o are the conditional model laws after record o. The joint laws $s(dm)K(m, do)$ and $t(dm)K(m, do)$ have divergence $D_{\mathrm{KL}}(s\|t)$ because their common kernel cancels. Factoring the same joint laws by their record marginal gives the displayed split. The last term is the distinction left unresolved by the record. A constant evaluator discards every distinction; a sufficient record leaves none.

For a model link Θ_{ij}^s, compatibility means $K_i(\Theta_{ij}^s m, A) = K_j(m, A)$ after identifying record alphabets. Only then does transporting the model law also transport its predictive meaning. If the link changes with the laws, this condition must continue to hold at the updated state.

5.3 Compression should name what matters

Rate–distortion theory makes the desired fidelity part of the communication problem [11]. A low-dimensional real vector alone is not a finite-bit channel. Noise, precision, or an encoding constraint must specify the communication cost. Wyner–Ziv coding additionally gives the receiver a realized variable correlated with the source [15]; possession of a prior by itself is not that realized side information.

For a predictive criterion, name the source X, target response Y, and accessible receiver information Z. An encoder generates U from X alone, so $U - X - (Y, Z)$ is a conditional-independence chain. The optimal decoder is $P(Y \mid U, Z)$. Its expected relative-entropy distortion is

$$\mathcal{D} = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{5.7}$$

The equality follows by expanding $I(Y; X, U \mid Z)$ in both orders. It is the information about the chosen response that compression discards. An information-bottleneck criterion trades this quantity against a cost such as $I(X; U \mid Z)$ [12]; an operational bit rate requires additional coding assumptions.

If $X = (A, B)$ consists of independent fair bits and $U = A$, the distortion is zero for target $Y = A$ and $\log 2$ for target $Y = B$. For a prospective parent, the target can be a future boundary response. Retaining one such response does not ensure that every later interaction can be inherited. That stronger requirement concerns dynamics.

Chapter 6

When a Collective Can Act as an Agent

A collective can have a simple description without being able to evolve from that description alone. Knowing the total energy of two bodies, for example, does not usually determine their next interaction. A parent of law-valued agents faces the same question: does its retained state determine its future responses?

There are two operations to distinguish. A channel C compresses represented contents and gives a parent law $Q_x C$. A map R retains a complete dynamical state $R(x)$. Besides belief and model laws, that state can include the model evaluator, source information, boundary channels, and memory. A distribution over possible contents does not automatically contain these additional data.

6.1 Two routes to the same parent

Suppose the fine configuration advances by F. Exact inherited evolution requires a parent update F^c such that

$$R(F(x)) = F^c(R(x)). \tag{6.1}$$

We can first evolve the children and then form their parent, or first form the parent and evolve it. Both routes must give the same retained state. Repeating the identity gives agreement at every later step. Boundary response mechanisms must likewise depend only on the retained state.

The immediate test is whether configurations with the same parent have the same next parent. If $R(x) = R(x')$ but $R(F(x)) \neq R(F(x'))$, no single F^c can satisfy the equation. Conversely, in a finite state space, this test is sufficient: evolve any representative of each parent state. Agreement within each fiber of R makes the choice irrelevant. On general measurable spaces the resulting map must also be measurable.

For stochastic dynamics, replace the next state by its distribution. If T is the fine transition matrix, the probability

$$\sum_{y:R(y)=z'} T(x, y) \tag{6.2}$$

must depend only on $R(x)$. These sums then define normalized parent rows. Conditioning on any parent history merely averages identical rows, so the parent reproduces the whole retained history law. This is strong lumpability [8]. Agreement for one equilibrium preparation is a weaker statement.

6.2 When a gradient flow descends

Suppose the fine configuration lies on a Riemannian manifold with metric g and evolves by $-\operatorname{grad}_g E$. Let a smooth surjective submersion R retain the parent, whose metric is g_c. One sufficient condition

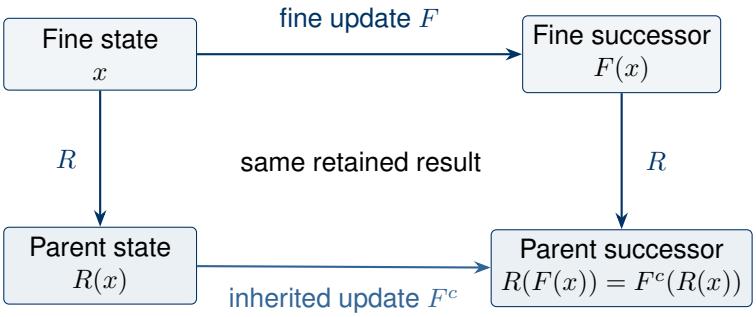


Figure 6.1: The two routes to the next parent configuration must agree. Vertical maps retain information; horizontal maps advance the dynamics. A compressed law alone need not make this square commute.

is that the energy depend only on the parent,

$$E = E_c \circ R, \tag{6.3}$$

and that R preserve lengths of horizontal vectors, meaning vectors orthogonal to $\ker DR$. Then the gradient is horizontal because the energy has no derivative along a fiber of R. For a horizontal u,

$$g_c(DR\,\mathrm{grad}_g\,E, DRu) = g(\mathrm{grad}_g\,E, u) = dE_c(DRu). \tag{6.4}$$

Every parent tangent has a horizontal lift, so the pushed gradient is exactly $\mathrm{grad}_{g_c}\,E_c$. The dynamics descends in the same time.

If instead $E = \chi(E_c \circ R)$ and the horizontal metric is scaled by $d(x)^2$, the pushed vector field acquires the factor $\chi'(E_c(Rx))d(x)^2$. A positive factor preserves oriented trajectories after a time change. A factor that depends on discarded state does not specify an autonomous parent clock. Taking the minimum of E over each fiber does not supply the first hypothesis: it gives the value at the minimizing states, not the fine energy everywhere on that fiber.

6.3 A missing role can prevent closure

Let two independent represented bits have laws $\mathrm{Ber}(p_1)$ and $\mathrm{Ber}(p_2)$. Their probabilities evolve according to

$$\begin{aligned} \dot{p}_1 &= -p_1 + \lambda(p_2 - p_1), \\ \dot{p}_2 &= -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \end{aligned} \tag{6.5}$$

The square $[0,1]^2$ is preserved: the derivative points inward at every boundary. Retain only the law of the sum of the two bits. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for sums $0, 1, 2$.

Their subsequent changes differ. The coupling terms cancel when we add the two equations, giving

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2). \tag{6.6}$$

The derivatives are -1.2 and -1.8. Thus the same retained law has two different possible derivatives. Compression erased which probability belonged to the faster-decaying role. The fine recognition law remains a product throughout. What must be retained is the information this evolution uses; cross-agent covariance is not automatically required.

6.4 An exact supplied hierarchy

A finite construction shows that two successive reductions can work. Take $x = (a, b, c, r) \in \{0, 1\}^4$, with internal register r and readout $\rho(x) = r$. Let the agent laws be

$$q_a = \mathrm{Ber}\left(\frac{1 + 2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2 + a}{4}\right). \tag{6.7}$$

The model evaluator assigns probability $1/4 + m/2$ to record value one when the model presentation is $m \in \{0, 1\}$. Averaging over s_a gives the predictive law $\nu_a = \mathrm{Ber}(1/2 + a/8)$.

At each step draw independent bits ξ, η, υ with probabilities $1/4, 1/3, 1/5$ of being one, independently of previous draws, and set

$$\begin{aligned} a' &= a \oplus \xi, \\ b' &= b \oplus a \oplus \eta, \\ c' &= c \oplus b \oplus \upsilon. \end{aligned} \tag{6.8}$$

Here $\oplus$ means addition modulo two. Independently of these draws, conditional on the current state, give r' law ν_a. The record is therefore an internal state change. Its generation rule is supplied, rather than deduced from the existence of an evaluator.

First retain (a, b, r), then retain (a, r). Summing the fine transition over c' removes a normalized factor and leaves a row independent of c. Summing that row over b' removes another normalized factor and leaves a row independent of b. The configuration spaces shrink from sixteen to eight to four states. Both parents retain q_a, s_a, their evaluator, and the register. They reproduce the complete retained state and record histories for every initial law. The coupling is real, since a affects b' and b affects c'; its triangular form permits these reductions. The hierarchy itself was prescribed.

6.5 Memory and the remaining formation problem

When closure fails, a parent can retain more state or, under a declared stochastic process, the conditional law of hidden states given its accessible history. In a finite model this is a probability vector: propagate it with the fine transition, restrict to the new retained value, and normalize. That vector determines every future response even when the current summary does not. Keeping only an equilibrium hidden distribution generally changes transient predictions.

There is an exact way to separate insufficient memory from inaccurate prediction. Let P be a declared law of a finite record sequence $Z_0, \ldots, Z_{T-1}$ on standard Borel spaces. A retained statistic is $M_t = f_t(Z_{<t})$. Let S use normalized prediction rows $s_t(\cdot \mid M_t)$. The path-law chain rule and

conditional disintegration give

$$D_{\mathrm{KL}}(P\|S) = \sum_{t=0}^{T-1} I_P(Z_t; Z_{<t} \mid M_t) + \sum_{t=0}^{T-1} \mathbb{E}_P D_{\mathrm{KL}}\big(P(Z_t \in \cdot \mid M_t)\|s_t(\cdot \mid M_t)\big). \tag{6.9}$$

The first sum is information about the next record that was discarded with the history. The second is error in the chosen prediction given the retained information. Both are nonnegative, including when the total is infinite. This extends the finite-window Markov decomposition [4]. Zero first sum gives predictive sufficiency for the declared process; a compact recursively updated memory still needs a realization $M_{t+1} = U(M_t, Z_t)$. Changing the allowed controls changes which histories must be predicted.

Statistical dependence alone also does not select an agent hierarchy. A declared finite cross-agent joint law has a finest partition into independent blocks, but arbitrarily weak couplings can merge those blocks. A deterministic selector reading only a fully permutation-symmetric law cannot choose a nontrivial intermediate hard partition while respecting relabeling: swapping one member with an outside agent would create a different block that overlaps the original. Realized histories, additional geometry, or a distribution over partitions can fall outside that restriction. The result identifies what a selector must read; it does not identify independence with agency.

Formation requires dynamics that choose persistent blocks, hand their laws and interactions to parents, and then form another collective from at least two such parents. The next interaction must be inherited, without resetting source data or adding a newly tuned attraction. The next chapter supplies one such fusion mechanism using a nonsmooth interaction. Its additional assumptions distinguish that example from the smooth KL dynamics discussed earlier.

Chapter 7

A Nonsmooth Route to Recurrent Formation

Smooth attraction can bring two laws progressively closer without making them equal at any finite time. An absolute-difference interaction behaves differently: its force does not vanish as the gap closes. This chapter gives a constitutive extension in which four agents form two parents, and those parents subsequently fuse under the inherited interaction. The grouping follows the evolving state; the equations contain no prescribed pairs.

Pairwise convex fusion penalties and finite-time nonsmooth consensus have established precedents [3, 6]. Here the additional task is to inherit the probability laws, model evaluator, metric, and subsequent responses when fusion occurs.

7.1 Laws, a model, and inherited mass

Write $\mathrm{LN}(\mu, \sigma^2)$ for the law of a positive random variable whose logarithm is normal with mean μ and variance σ^2. A component containing n primitive agents carries

$$q_n = \mathrm{LN}(\mu^q, 1/n), \qquad s_n = \mathrm{LN}(\mu^s, 1/n), \qquad n \geq 1. \tag{7.1}$$

Recognition is the product $q_n \otimes s_n$. Its normalized reference model p_n is jointly Gaussian in the logarithms, with zero mean and precision

$$n \begin{pmatrix} a & -c \\ -c & a \end{pmatrix}, \qquad a > c > 0. \tag{7.2}$$

Thus the model couples the two sectors even though recognition is factorized. Its normalized conditional evaluator is

$$G_n(m, dk) = \mathrm{LN}\left(\frac{c}{a}\log m, \frac{1}{an}\right)(dk). \tag{7.3}$$

Direct Gaussian relative entropy gives

$$D_{\mathrm{KL}}(q_n \otimes s_n \| p_n) = C + nU, \qquad U = \frac{a}{2}\left[(\mu^q)^2 + (\mu^s)^2\right] - c\mu^q\mu^s, \tag{7.4}$$

where $C = a - 1 - \frac{1}{2}\log(a^2 - c^2)$. The mixed term makes the belief update depend on the model coordinate and conversely. The Fisher metric of this recognition family is

$$ds^2 = n\left[(d\mu^q)^2 + (d\mu^s)^2\right]. \tag{7.5}$$

Both the metric and the local potential therefore carry the same inherited mass.

A parent communicates at primitive resolution through the normalized channel

$$B_n(k, dw) = \mathrm{LN}(\log k, 1 - 1/n)(dw). \tag{7.6}$$

At $n = 1$ this means the identity channel. Adding the independent log variances shows that $q_n B_n = \mathrm{LN}(\mu^q, 1)$, with the analogous identity for s_n. This restores a primitive message law while retaining the narrower parent law internally.

7.2 The interaction that permits contact

All means here use common reference coordinates. In sector $\sigma \in \{q, s\}$, transform the source's restored message by

$$S^\sigma_{IJ}(w) = \exp\big[\chi_\sigma(\mu^\sigma_I - \mu^\sigma_J)\big]w, \qquad 0 < \chi_\sigma < 1. \tag{7.7}$$

Its log mean moves toward the receiver. Including endpoint frames U^σ_I, U^σ_J gives the displayed link $U^\sigma_I \exp[\chi_\sigma(\mu^\sigma_I - \mu^\sigma_J)](U^\sigma_J)^{-1}$, with the usual endpoint covariance under passive re-expression. The interaction depends on the laws, rather than only identifying frames.

The compared message variances are both one. Consequently,

$$\sqrt{2D_{\mathrm{KL}}\left(\widehat{\nu}^\sigma_I \| (S^\sigma_{IJ})_\# \widehat{\nu}^\sigma_J\right)} = (1 - \chi_\sigma)|\mu^\sigma_I - \mu^\sigma_J|, \tag{7.8}$$

where the displayed comparison is in a common reference presentation and $\widehat{\nu}$ denotes a restored law. Rescaling the coefficient by $1/(1-\chi_\sigma)$ gives an absolute-difference penalty with strength $\lambda > 0$. The square root is the new constitutive choice. It changes the smooth quadratic peer penalty into a convex function with a corner at equality.

On the complete contact graph, the energy apart from constants is

$$\mathcal{E} = \sum_I n_I U_I + \lambda \sum_{I<J} n_I n_J \left(|\mu^q_I - \mu^q_J| + |\mu^s_I - \mu^s_J|\right). \tag{7.9}$$

The coefficient $n_I n_J$ counts primitive contacts. With unit Fisher mobility the evolution is $n_I \dot{\mu}_I \in -\partial_{\mu_I}\mathcal{E}$. The subgradient of $|x|$ is its sign away from zero and the interval $[-1, 1]$ at zero.

Convexity makes this evolution unique. For two solutions, the derivative of their squared distance in the fixed mass metric is the negative pairing of the state difference with a subgradient difference, hence is nonpositive. Coincident primitive agents are interchangeable in the complete-graph equations. Uniqueness then keeps them equal after contact. Sticking follows from the same dynamics that brought them together.

7.3 Two successive formation events

Consider the invariant family

$$\mu_i^q = u_i + v, \qquad \mu_i^s = u_i - v, \qquad v(t) = v_0 e^{-(a+c)t}. \tag{7.10}$$

Both sectors have the same interagent gaps, so their interaction forces cancel in the equation for v. Set $\kappa = a - c > 0$ and introduce

$$u_i(t) = e^{-\kappa t} z_i(\tau), \qquad \tau = \frac{e^{\kappa t} - 1}{\kappa}. \tag{7.11}$$

This removes the common linear contraction. Before collisions,

$$\frac{dz_I}{d\tau} = -\lambda \sum_{J \neq I} n_J \operatorname{sign}(z_I - z_J). \tag{7.12}$$

Start four primitive agents at $z = (-3d, -2d, 2d, 3d)$, with $d > 0$. Their slopes are $(3\lambda, \lambda, -\lambda, -3\lambda)$. Each neighboring outer pair closes a gap d at relative speed 2λ. The resulting stages are

Time	Positions in z	Masses
0	$-3d, -2d, 2d, 3d$	$1, 1, 1, 1$
$\tau_1 = d/(2\lambda)$	$-3d/2, 3d/2$	$2, 2$
$\tau_2 = 5d/(4\lambda)$	0	4

After the first event, the parents have slopes 2λ and -2λ. Their remaining gap $3d$ closes in another $3d/(4\lambda)$. Physical event times are $t_r = \kappa^{-1} \log(1 + \kappa\tau_r)$. No coupling or mobility is retuned at the second level.

7.4 What the parent retains

At synchronization, independent log samples with a common mean have an average of variance $1/n$. Exponentiating that average gives the geometric-mean retention and the stated parent law. Restricting the primitive Fisher metric to equal location coordinates gives the factor n. Summing contact forces gives $n_I n_J$, so the reduced dynamics use exactly the inherited metric and contacts. Redundant location coordinates are removed after synchronization; an average of separated children is not claimed to close their dynamics.

Individual sample distinctions require care. If X is the retained log average, the entire primitive vector is reconstructed as

$$(X_1, \ldots, X_n) = X\mathbf{1} + R, \qquad R \sim N\left(0, I - \frac{\mathbf{1}\mathbf{1}^T}{n}\right), \tag{7.13}$$

with R independent of X. The residuals sum to zero and are correlated. Independent scalar applications of B_n to one reused parent sample preserve individual marginals but give the wrong joint law.

Fresh record sampling also closes explicitly. Applying G_n to s_n gives

$$\mathrm{LN}\left(\frac{c}{a}\mu^s, \frac{(c/a)^2 + 1/a}{n}\right), \tag{7.14}$$

the law of the geometric mean of n fresh primitive model records. A primitive record instead uses a fresh restored model sample followed by G_1. These are exact statements for the declared fresh sampling protocols; they do not establish closure for sampled-record feedback into the dynamics.

This extension establishes recurrent fusion on the common-v invariant family. It produces eventual global consensus, rather than persistent community diversity. The square-root penalty is a constitutive addition, formation under the original smooth KL dynamics remains open, and no geometry of the context base is derived.

Chapter 8

Keeping Distinct Children Under Smooth Dynamics

Exact inheritance need not identify the children with one another. The requirement is that discarded distinctions have no effect on the future quantities the parent promises to reproduce. Under smooth KL dynamics this requirement has a concrete content: hidden conditional information can contribute to the force on a retained marginal. A parent must retain that contribution, or have a reason why it cancels.

8.1 A hidden conditional law can move a visible marginal

Write a law as $q(dz, dv) = \mu(dz)\kappa_z(dv)$. Here z is retained and v is internal. Against a reference $\nu(dz)A(dv)$, the KL chain rule is

$$D_{\mathrm{KL}}(q\|\nu \otimes A) = D_{\mathrm{KL}}(\mu\|\nu) + \int H(z)\mu(dz), \qquad H(z) = D_{\mathrm{KL}}(\kappa_z\|A). \tag{8.1}$$

If one public value has a more costly hidden conditional law, changing its probability changes the full energy. The public Fisher velocity therefore contains $H(z)$ as well as the ordinary public force. A single average hidden entropy does not tell the parent which public values carry that cost.

There is an exact architecture in which this is the only hidden contribution. A sender transmits its public coordinate through a normalized, possibly law-dependent channel. The receiver reconstructs its private coordinate from its own conditional $\kappa_{i,z}$. The receiver law and compared source law then have the same private conditional, so that conditional cancels from their peer KL. With a fixed reference whose private factors are independent, the two-sector local model still couples the public belief and model laws. The original potential and Fisher metric give

$$\begin{aligned} \dot{\mu} &= -\mu[h^c + H - \mathbb{E}_\mu(h^c + H)], \\ \dot{\ell}_z(v) &= H(z) - \ell_z(v), \qquad \ell_z = \log(d\kappa_z/dA). \end{aligned} \tag{8.2}$$

The public score h^c includes every incoming interaction and every channel derivative. Cancellation is a functional identity, so it does not require freezing a live channel while differentiating.

8.2 Retain the distribution of density values

The conditional equation in eq. (8.2) moves a log-density value according to its current value and the conditional KL. It does not need the name of the private point carrying that value. This suggests

retaining the probability law

$$\xi_z = (\ell_z)_{\#}A, \qquad \int e^r \xi_z(dr) = 1, \qquad H(z) = \int r e^r \xi_z(dr). \tag{8.3}$$

For example, use a uniform reference on two private states. The laws $(3/4, 1/4)$ and $(1/4, 3/4)$ have the same spectrum: half the reference mass sees $\log(3/2)$ and half sees $\log(1/2)$. Their private preferences differ, but their contribution to the public force agrees.

The complete spectrum evolves exactly. With $a = e^{-t}$,

$$\xi_z(t) = \left(r \mapsto ar - \log \int e^{ar'} \xi_z(0, dr') \right)_{\#} \xi_z(0). \tag{8.4}$$

This follows by solving $d\kappa_z(t)/dA \propto (d\kappa_z(0)/dA)^a$ and normalizing. The formula applies to arbitrary dominated laws under the stated regularity conditions; it does not fit a Gaussian. The retained state can remain infinite-dimensional.

If all public values within an agent have the same conditional spectrum, $H(z)$ is constant and disappears under Fisher centering. The public laws then follow the original coarse KL dynamics exactly. The private conditionals can still differ: a public bit can select between the two swapped binary laws above. This gives internal correlation without changing the public force. Other agents may have different public laws and different spectra; equality between agents is unnecessary.

This property is not created at a finite time by the conditional relaxation. At every finite time $a > 0$, the power transformation and its normalization can be inverted. Different spectra therefore remain different until an asymptotic limit, unless some additional operation changes them. Retaining unequal spectra is an exact alternative, but their existence alone does not select groups of agents.

8.3 What a future record is allowed to inspect

The same two swapped private laws show why a boundary protocol matters. Let record 1 have probabilities $(3/4, 1/4)$ at the two private points. The record probabilities are respectively $5/8$ and $3/8$. Equal unmarked spectra therefore do not preserve this response.

If all permitted private readouts use a fixed mark $T(v)$, retain the joint reference distribution of $(\ell_z(v), T(v))$. A likelihood $L(o \mid z, T(v))$ then gives the conditional record probability

$$N_z(o) = \int e^r L(o \mid z, d) \eta_z(dr, dd), \qquad \eta_z = (\ell_z, T)_{\#}A. \tag{8.5}$$

After the record, the log-density coordinate changes by

$$r^+ = r + \log L(o \mid z, d) - \log N_z(o). \tag{8.6}$$

The public marginal is multiplied by $N_z(o)$ and normalized. Between records the same smooth conditional relaxation resumes. This gives an exact inherited record-and-update sequence for the declared protocols. If the marks identify every private point, however, the supposed compression may retain everything. Sufficiency is relative to what the future interaction can ask.

8.4 A smooth channel can erase a dependence in finite time

A separate four-agent example shows that exact independence can become available during the original smooth flow. Each belief and model law has a public bit, a private bit, and an internal gate bit. A fixed normalized local evaluator couples the public belief and model means. The gate means relax toward a fixed reference value. The private part of a peer channel depends on the source's private mean multiplied by a smooth function α of the population's gate means.

The function α has an exactly flat zero region. Its derivative is included in the Fisher equations. By choosing its private gain sufficiently small, the local KL force drives the gate means into this region at a finite time and keeps them there. Once inside, every peer's private cost in a receiver row is the same for all sources. It cancels from the Gibbs source comparison. The reduced energy and Fisher metric then split into a public part and an independent private-and-gate part.

The public states start in two invariant sign regions, with two agents in each. No labels are assigned to groups by the equations. Initially, private similarities can select crossed pairs, each containing one member from each public region. After private erasure, the preferred sources change: each agent prefers the other member of its public region, and those pairs persist. All cross-pair source weights remain positive. The children need not coincide; smooth uniqueness keeps distinct public states distinct at every finite subsequent time. Their separation is not claimed to stay bounded below forever.

For each pair, retain the empirical measure of its two paired public belief and model means. The original public vector field gives its exact evolution, and fresh public responses are computed by selecting and sampling a child from that measure. The private and gate laws can be forgotten after erasure. This is a noninjective reduction, but the empirical measure still retains every public child state, up to its label.

The example is an existence construction with an engineered channel, a product preparation on finite alphabets, and a specified class of boundary responses. It adds no fusion penalty and imposes no synchronization. It does not show that arbitrary smooth channels acquire a sufficient interface, that the distinct public child states admit a smaller aggregate description, or that the two resulting parents form another parent. Those are the remaining formation obligations.

Chapter 9

A collective trajectory is not a complete interface

There is a way to compress distinct public laws under the original smooth interaction. It uses their collective motion. Near a common stable reference, differences between agents can decay faster than their shared evolution. One can then assign each fine configuration a virtual collective state whose future follows exactly the same reduced trajectory. The fine agents need not coincide at any finite time.

For an explicit general-law construction, take probability references A, B with balanced binary features φ, ψ, and the fixed normalized evaluator

$$G(m, dk) = \frac{e^{J\varphi(k)\psi(m)}}{\cosh J} A(dk), \qquad |J| < 1/4. \tag{9.1}$$

The other sample coordinates can be arbitrary. Write $q_i = e^{u_i} A / \mathbb{E}_A e^{u_i}$ and $s_i = e^{v_i} B / \mathbb{E}_B e^{v_i}$, with bounded reference-centered log densities. No finite-dimensional family is imposed. Use identity channels, uniform nonself source priors, and the original Gibbs-reduced KL potential.

Set $z_i = (u_i, v_i)$. On the virtual diagonal the common law follows

$$\dot{u} = -u + J\varphi \mathbb{E}_{s_v} \psi, \qquad \dot{v} = -v + J\psi \mathbb{E}_{q_u} \varphi. \tag{9.2}$$

At the reference state, the linearized fine dynamics is

$$\dot{z}_i = \mathcal{A} z_i - \frac{2N}{N-1}(z_i - \bar{z}), \qquad \bar{z} = \frac{1}{N} \sum_i z_i. \tag{9.3}$$

Here $\mathcal{A}$ is the linearization of eq. (9.2). Its slowest common decay rate is $1 - |J|$, whereas the slowest difference rate is $1 + 2N/(N-1) - |J|$. The factor two in the peer term includes both receiving and being used as a source.

The arithmetic average of the log densities is not itself an exact closed state. Away from the diagonal, its equation contains a correction quadratic in the child differences. That correction decays fast enough to define a nonlinear phase. Let Φ_t be the fine flow and ϕ_t the flow of eq. (9.2). The exact parent coordinate is

$$\Pi(x) = \lim_{t \to \infty} \phi_{-t}\big(\overline{\Phi_t x}\big), \qquad \Pi(\Phi_t x) = \phi_t(\Pi x). \tag{9.4}$$

The backward common flow transports the late collective state back to the initial time. The fast decay of child differences makes the limit converge. Its local fibers contain continuously many

distinct child configurations, so this construction discards public degrees of freedom rather than merely their labels. The parent evolves exactly once its initial phase is known. Evaluating that phase remains an implicit dynamical problem.

The response of the group is a separate question. For two binary agents with equal belief and model means within each agent, put $x_1 = m + a$ and $x_2 = m - a$. Their phase-law mean is

$$p = m + \frac{3(2-J)}{2(5-J)} m a^2 + o((|m| + |a|)^3). \tag{9.5}$$

A uniform choice of child followed by one bit sample has mean m. The diagonal configuration with both child means equal to p has the same phase parent, but its corresponding response has mean p. For small $m = a > 0$, these means differ. The parent predicts its own inherited trajectory while failing to reproduce this ordinary group response. Sampling the phase law would change the interface.

An unrestricted identity-channel Gibbs receiver can distinguish even more. For child densities f_j and positive weights w_j, its block normalizer is

$$Z(r) = \sum_j w_j e^{-D_{\mathrm{KL}}(r\lambda \| f_j \lambda)/T}. \tag{9.6}$$

Knowing this response for all nearby receiver laws determines every distinct exposed child density and its total weight. To see the mechanism, cancel the known receiver entropy. The remaining function is a finite sum of exponentials of the linear probes $\int r \log f_j$. A direction that separates these probes gives a Vandermonde system, which recovers their coefficients. Distinct child laws therefore cannot be forgotten while retaining this entire interface.

Restricting the possible receivers may permit compression. It must be accompanied by an exact check of the incoming forces and of repeated-response memory. For a second formation step, joining parents must also respect the information already discarded. The collective phase supplies a continuous autonomous reduction without an erasure gate. It does not yet supply those recursive interface properties or a dynamically selected hierarchy.

Chapter 10

A Parent That Keeps Its Responses

A parent can now be constructed that preserves both evolution and responses, through any finite number of supplied binary levels. The children keep distinct public laws, the evaluator couples belief and model, and the original smooth identity-channel KL interaction is unchanged. The construction works because a particular class of public patterns is invisible to fresh uniform-descendant sampling, and remains invisible under the inherited evolution. It does not show the dynamics discovering that class.

10.1 Four children and three public circles

Take a general base variable r with reference law A, and three independent public angles. Each angle determines a balanced sign, for example

$$\sigma_A = \operatorname{sign}\cos(\theta_A - \alpha_A(r)), \tag{10.1}$$

and similarly σ_B, σ_C. The phase $\alpha_A(r)$ can be any measurable function. Rotating it changes which actual public samples carry positive or negative contrast.

Choose positive profiles $m_+(r), m_-(r)$ and contrast magnitudes $0 < a_+(r), a_-(r) < 1$, with $\int (m_+ + m_-)dA/2 = 1$. Four child densities are

$$\begin{aligned} q_{A\pm} &= m_{\sigma_C}(1 \pm a_{\sigma_C}\sigma_A), \\ q_{B\pm} &= m_{-\sigma_C}(1 \pm a_{-\sigma_C}\sigma_B). \end{aligned} \tag{10.2}$$

They are laws on the full base-and-angle space. There is a separate copy for the model laws. A fixed joint reference $\rho(r, w)$ couples the belief and model bases and is independent of the angles. Every child still compares its full law with every other child through identity channels.

If a response freshly chooses $A+$ or $A-$ with equal probability and then samples that child, its density is m_{σ_C}. The sign σ_A cancels pointwise. Its phase therefore changes the two individual public laws without changing this pair response. The pair nevertheless retains $a^2_{\sigma_C}$: its magnitude influences the future mixture through the KL forces.

The two first parents have response densities m_{σ_C} and $m_{-\sigma_C}$. They are distinct when $m_+ \neq m_-$. Choosing one parent uniformly cancels σ_C and gives the full-coordinate response density

$$\bar{m}(r) = \frac{m_+(r) + m_-(r)}{2}. \tag{10.3}$$

The root also retains the unordered joint pair

$$\left\{(m_+(r), a_+(r)^2), (m_-(r), a_-(r)^2)\right\}. \tag{10.4}$$

It can discard the phase of σ_C while retaining the magnitudes that determine its future. A density value and its contrast must remain paired.

10.2 Why the original dynamics closes

For a tree of depth d, assign a balanced public sign to every internal node. A leaf reads one sign on each step of its path. Write the resulting bit string as $u \in \{-1, 1\}^d$, and let its density be $f(r, u)$. This profile is an arbitrary positive bounded function; it is not required to remain a product of independent factors.

Two leaves that separate immediately after their first k common edges have the same first k bits and opposite next bits. Their later signs come from separate branches. Define P_k to keep the shared bits, flip the next bit, and average the remaining bits. The exact original peer cost is

$$D_k(f) = \int f[\log f - P_k \log f] dA dU_d, \tag{10.5}$$

where U_d is the uniform bit law. The source classes have evolving total weights b_k. Differentiating the complete energy gives

$$h = 2\log f + V - \sum_k b_k \left[P_k \log f + \frac{P_k f}{f}\right], \qquad \dot{f} = -f(h - \mathbb{E}_f h). \tag{10.6}$$

Here $V(r) = -\int g_0(w) \log \rho(r, w) dB(w)$ carries the other sector's base marginal. The ratio $P_k f/f$ comes from being a source in other agents' costs. Omitting it would give a different flow.

All quantities in eq. (10.6) use the profile values and their nested grouping. A branch exchange preserves them. At every physical sample point, the leaves below a node run through every possible suffix bit string, so averaging their forces produces exactly the same profile equation. Both the original energy and the original Fisher metric carry the factor 2^d; it cancels in the inherited equation. There is no inserted coarse timescale.

The parent therefore keeps an unordered tree of profile values. Joining siblings places their two retained trees inside one unordered pair. Subsequent joins can be evaluated in the same way, and the result agrees with retaining the entire subtree directly. The grouping must remain: a flat multiset would forget which profiles enter each D_k. The parent state can still grow with the number of children.

This joining changes the description of one already coupled fine system. It does not switch on new edges or prove that the agents have selected that tree. Original uniform source rows are allowed; their class totals evolve under the original row equations.

10.3 What the inherited interface permits

At every level, choose a descendant freshly and uniformly, draw its full original sample, and conceal its label. The retained subtree mean gives the exact response density. Choosing a parent and then a descendant gives the same response as direct descendant selection. Repeat with fresh labels and fresh samples, using independent labels for belief and model responses.

Feedback also closes for common record likelihoods depending on the base. Every child has the same base marginal $f_0(r)$, so a record with likelihood $L_e(o \mid r)$ has the common normalization $Z_o = \int f_0 L_e dA$. Its exact component update is

$$f^+(r,u) = \frac{L_e(o \mid r) f(r,u)}{Z_o}. \tag{10.7}$$

Every value at that base point receives the same multiplier, which commutes with every tree retention. The next experiment may depend on the previous record history. Induction then preserves the full finite history law, with the original continuous dynamics between updates.

Reusing one hidden child for two samples changes the response. For a pair $q_\pm = m(1 \pm a\sigma)$ it gives

$$m(k)m(k')[1 + a(k)a(k')\sigma(k)\sigma(k')]. \tag{10.8}$$

The cross term reads the discarded public pattern. Unrestricted external probes can also expose it. Exact inheritance is a statement about the declared interface, not every experiment one could perform on the children.

Finally, a phase rotation changes the public laws continuously in total variation: the angular region whose sign changes is proportional to the rotation angle. The corresponding squared Hellinger distance has the same linear dependence, so these phase paths have no finite Fisher tangent. The original time evolution remains smooth on each fixed-partition profile chart. This construction establishes recursive boundary inheritance on a prepared family. Obtaining that organization from less structured initial conditions is the next formation problem.

Appendix A

Notation in Physical Terms

Symbol	Meaning and role
k, m	A represented state possibility and a model presentation. Neither symbol by itself denotes a probability law.
q_i, s_i	The agent's belief law and model law. Their marginals can evolve in coupled equations.
ζ_i	The joint law of represented state and model contents; $q_i \otimes s_i$ is a declared product restriction.
g^F, ℓ_v	Fisher distinguishability on a regular statistical family and the score describing one infinitesimal change.
$D^\omega \zeta, h^\omega_\zeta$	Statistical change after contextual transport is accounted for, and the resulting tensor on the context base.
$A^F, \mathfrak{k}$	The collectively minimizing connection class and the metric measuring common infinitesimal frame motions.
$G_i(m, dk)$	An evaluator assigning state predictions to each model presentation. Averaging gives $s_i G_i$.
$x \in \mathsf{S}$	A complete configuration of agents, including their laws and the memory and interfaces read by the dynamics.
$\rho_i : \mathsf{S}_i \to \mathsf{O}_i$	A readout of an internal distinction. Its value is a record.
Γ_{ij}	A normalized interaction transition on complete configurations. It induces record probabilities through the readout.
U^a_i	An invertible frame map for sector $a = q, s$. Changing frame changes the description.
T^a_j, O^a_i	Sending and receiving interfaces. Their product can discard directions.
L^a_{ij}, Θ^a_{ij}	A reference generator and its framed exponential interaction. The generator may be singular while its exponential is invertible.
β_{ij}, γ_{ij}	Belief-sector and model-sector source weights in normalized rows.
$\mathcal{E}$	A declared interaction potential on agent laws. Its gradient dynamics requires a metric.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model and conditioned record.
$Q_x C$	A law of compressed represented contents. It does not specify a parent update.
$R(x), F^c$	A complete retained state and its inherited update. They must satisfy $RF = F^c R$.
$\mathcal{C}, B, F$	The context base, a connection one-form, and its curvature. These require a rule for relating neighboring contexts. The symbol F also denotes a discrete update in the inheritance chapter.

Bibliography

[1] Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2):251–276, 1998. doi: 10.1162/089976698300017746.

[2] Nihat Ay, Jürgen Jost, Hông Vân Lê, and Lorenz Schwachhöfer. Parametrized measure models. *Bernoulli*, 24(3):1692–1725, 2018. doi: 10.3150/16-BEJ910. URL https://arxiv.org/abs/1510.07305.

[3] Jorge Cortés. Finite-time convergent gradient flows with applications to network consensus. *Automatica*, 42(11):1993–2000, 2006. doi: 10.1016/j.automatica.2006.06.015.

[4] Robert M. Gray. *Entropy and Information Theory*. Springer-Verlag, New York, 1990. URL https://www-ee.stanford.edu/~gray/it.pdf. Corrected first edition, revised June 26, 2023.

[5] A. Guichardet. On rotation and vibration motions of molecules. *Annales de l'I.H.P. Physique théorique*, 40(3):329–342, 1984. URL https://www.numdam.org/item/AIHPA_1984__40_3_329_0/.

[6] Toby Dylan Hocking, Armand Joulin, Francis Bach, and Jean-Philippe Vert. Clusterpath: An algorithm for clustering using convex fusion penalties. In *Proceedings of the 28th International Conference on Machine Learning*, pages 745–752, 2011. URL https://icml.cc/2011/papers/419_icmlpaper.pdf.

[7] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.

[8] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer, New York, 1976. ISBN 9780387901923. Reprint of the 1960 edition, with a new appendix.

[9] Debra Lewis, Nilima Nigam, and Peter Olver. Connections for general group actions, 2003. URL https://arxiv.org/abs/math/0309234.

[10] David Pollard. A note on insufficiency and the preservation of Fisher information. In *From Probability to Statistics and Back: High-Dimensional Models and Processes*, volume 9 of *IMS Collections*, pages 266–275. Institute of Mathematical Statistics, 2013. doi: 10.1214/12-IMSCOLL919. URL https://arxiv.org/abs/1107.3797.

[11] Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE International Convention Record*, 7(4):142–163, 1959. URL https://gwern.net/doc/cs/algorithm/information/1959-shannon.pdf.

[12] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method, 1999. URL https://arxiv.org/abs/physics/0004057. Manuscript dated September 30, 1999; arXiv:physics/0004057, posted 2000.

[13] David Tong. Gauge theory. University of Cambridge lecture notes, 2018. URL https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf.

[14] Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974. doi: 10.1103/PhysRevD.10.2445.

[15] Aaron D. Wyner and Jacob Ziv. The rate-distortion function for source coding with side information at the decoder. *IEEE Transactions on Information Theory*, 22(1):1–10, 1976. doi: 10.1109/TIT.1976.1055508.